LocalMind demo and speaking guide

A ten minute presentation updated 17 October 2026

Use this guide to explain LocalMind, run a prepared two minute demonstration and answer viva questions. The speaking pages match the current Markdown script. Read only speech paragraphs aloud; operation cues, preparation and optional answers are outside the ten minute timing.

Local-First Agentic Knowledge Copilot with a Custom MCP Memory Server

SIT | Minors in AIML | Academic year 23-27
Guide Dr. Sonali Kothari

Presenters and running order

Presenter	PRN	Time
Trijal Burman	23070123145	00:00-02:15
Tiya Singh	23070123144	02:15-04:30
Gayatri Rajesh Apotikar	23070123055	04:30-06:30
Nimita Jestin	23070122259	06:30-10:00

Speaking roles organize the presentation; they do not assert individual implementation contributions. Gayatri operates the app during the demonstration.

Find what you need

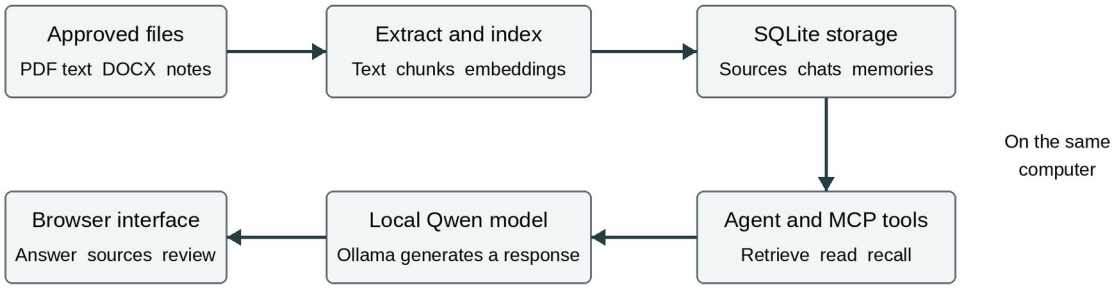
Pages	Use
2	Understand the system and explain RAG versus MCP
3-4	Prepare the app and follow the timed demonstration
5-8	Read the four verbatim speaking parts in order
9-11	Rehearse the 15 assigned viva answers
12	Check results, boundaries, and the final rehearsal

The ten minutes include the demo. Practise once with a timer; stop a slow model request after the demonstration wait limit and continue with the source and memory workflow.

Understand the project in plain language

LocalMind is a workspace for information already saved on a laptop. A user chooses folders on an HDD or SSD, indexes supported files, searches their contents, checks sources, and saves useful project context for later. The local model can help write an answer, summary, or draft plan, subject to its limits.

How a request moves through the system



The diagram shows the document workflow. The app handles greetings without a model or retrieval. File inventories use MCP index metadata. Document questions use the local API, SQLite evidence, the MCP stdio server and Ollama inference; the browser displays the result for review.

RAG and MCP solve different problems

RAG means retrieval augmented generation. Search finds relevant passages and supplies selected evidence with the question. Current records preserve the actual filename, path and document identifier. Cards group excerpts from one file while keeping original citation numbers. Retrieval and real references still do not guarantee a correct claim.

MCP means Model Context Protocol. It gives clients a common way to discover and call tools. Our server exposes local knowledge and memory functions. MCP does not rank passages by itself and is not a model or a vector database.

What memory means here

An explicit memory is a saved database record. A later request can retrieve it. Conversation history, extracted text, and plans are also stored locally. A model context window is temporary; saving a record does not train or fine-tune model weights, and every stored record is not sent on every turn.

The practical boundary

Initial software and model downloads require internet. Normal inference, indexing, and storage can then stay local without a required paid hosted API. Use a local MCP client throughout; a cloud-connected client could forward returned content. Local operation does not make generated answers automatically correct or encrypt the database.

Prepare the app before the clock starts

Prepare before the clock starts. Show only the two supplied sample files. Verify the chosen workspace is running; this update does not establish that the separate demo on port 8766 is ready.

Use a separate demo workspace

After normal Linux setup, run bash output/presentation/start-presentation-demo.sh from the checkout root. It uses separate sample data on port 8766 and the shared local model service. Preserve normal data and close private tabs.

Check local model readiness

In Settings, refresh runtime status and verify readiness. Source defaults remain Qwen3 4B / 8192. The optional Qwen3.5 9B / 12288 profile needs more memory. Warm the model beforehand; an HDD cold load may take a minute or longer.

Connect the samples

Open Library > Connect folder. Enter the absolute path to examples/knowledge, label it Lighthouse demo, then choose Connect & index. Wait until indexing finishes and check any errors.

Prepare the project conversation

Create Lighthouse Presentation in Projects > New project. Use Open project chat. Start a new conversation if its scope is wrong; the project selector is disabled in an existing session.

Verify the source and rehearse once

Read lighthouse-project.md at Storage decision. Rehearse the question on page 4 after the model is loaded. The ten second wait limit applies to this warmed demonstration. Keep the source reader available.

Useful navigation

Task	Click path
Find a passage	Library > Search library > Search > Content matches > Read document
Read or export text	Read document > Full extracted text or Download text
Save a note manually	Memory > Add memory > Memory, Type, Scope > Save memory
Review project memory	Memory > project filter > Lighthouse Presentation

Use examples/knowledge inside your checkout; on Windows use the actual checkout path followed by \examples\knowledge. The normal app address is http://127.0.0.1:8765. The Linux demo uses http://127.0.0.1:8766 when its launcher is running.

Run the two minute demonstration

Gayatri controls the keyboard from 04:30 to 06:30 of the full presentation.

Use these exact inputs

Question
What storage does the Lighthouse project use?

Explicit memory command
/remember Use only the supplied Lighthouse sample files in this presentation.

Expected source fact: SQLite. Verify it in lighthouse-project.md under Storage decision. This is a known sample fact, not a promise that the model will answer correctly.

The retained 17 October context-repair rehearsal returned SQLite with a source and preserved memory. That dated record was not repeated for the latest greeting and metadata fixes. Check the demo before presenting; prepared rehearsal evidence is not a fresh live result.

Follow the sequence

Demo time	Operate
00:00-00:25	Library > lighthouse-project.md > Read document. Show Storage decision. Say G1 from page 7.
00:25-00:55	Close the reader. Projects > Lighthouse Presentation > Open project chat. Send the question once. Wait at most 10 seconds. If ready, open the source card and compare it. Say G2.
00:55-01:35	Send the /remember command. After Saved, open Memory and select Lighthouse Presentation. Point to the note. Say G3.
01:35-02:00	Reload Memory once. Show the saved note, stop clicking, and hand over to Nimita with G4.

If the model is slow or wrong

At ten seconds, move on. Show the indexed source directly. If a chat is still busy, save the note using Memory > Add memory and select Lighthouse Presentation under Scope. Use the accurate fallback language on page 7; do not repeat attempts until an answer looks good.

A page reload demonstrates that a saved record survives that reload. It is not a live application restart test. Storage tests provide separate persistence evidence.

Keep these outside the timed demo

No new folder scans, downloads, application restarts, long summaries, plan generation, or extra question attempts. Summary and plan actions are optional extensions only if the panel explicitly allows more time.

Trijal Burman

Verbatim script | 00:00-02:15

Read the regular text aloud. Italic operation cues are silent.

Operate T1 Show the title or app welcome screen. Face the panel.

Good morning, respected panel members. We are Trijal Burman, Tiya Singh, Gayatri Rajesh Apotikar, and Nimita Jestin from SIT. Under the guidance of Dr. Sonali Kothari, we are presenting LocalMind: a Local-First Agentic Knowledge Copilot with a Custom MCP Memory Server.

Operate T2 Keep the problem statement visible. Do not begin navigating.

Our problem is scattered knowledge and lost project context. A useful decision may be in a document, a lecture note, or an earlier conversation. Finding it again takes effort, and an ordinary chat has a limited context window. We wanted a workspace that helps a person find, inspect, and reuse their own approved information.

Operate T3 Point to Library, Memory, and Projects.

The prototype searches supported local files, shows source references, reads extracted document text, saves explicit memories, and drafts project plans. It uses existing local models rather than training a new large model. A saved memory belongs in a database; a relevant passage is supplied to the model when a request is made.

Operate T4 Show the scope or privacy note.

Local-first does not mean unrestricted drive access. The user chooses folders on the HDD or SSD. Initial software and model downloads need internet, but the intended workflow can then run offline without a paid hosted API. Hardware, electricity, and storage still have costs. Local processing also does not guarantee that every generated statement is correct.

Operate T5 Show the architecture on page 2. Hand over to Tiya.

Tiya will now explain how retrieval, persistent storage, and our custom MCP server work together. The speaking roles we have assigned today are presentation roles, not a claim about each person's individual implementation contribution.

Tiya Singh

Verbatim script | 02:15-04:30

Read the regular text aloud. Italic operation cues are silent.

Operate Y1 Trace approved files, extraction, and SQLite in the figure.

The workflow starts with an approved folder. Supported text is extracted and divided into smaller overlapping passages. SQLite stores the complete extracted text, passage information, conversations, and memories. EmbeddingGemma represents passages as numerical vectors, while SQLite full-text search finds matching words. The retrieval layer combines these signals to select useful evidence.

Operate Y2 Point from retrieved evidence toward the model.

RAG means retrieval-augmented generation. We retrieve evidence and provide selected passages with the question to a language model. Qwen runs locally through Ollama. This laptop uses the optional Qwen3.5 nine-billion-parameter profile with twelve thousand two hundred eighty-eight context tokens. Source defaults remain four billion parameters and eight thousand one hundred ninety-two tokens. The paper measured the earlier model. References help us inspect evidence; they do not guarantee a correct answer.

Operate Y3 Point to the MCP tool connection.

MCP has a different purpose: it is a standard interface for discovering and calling tools. Our custom local server provides search, document reading, memory recall, and project functions through a real stdio connection. RAG is the information-finding approach; MCP is the tool connection. It is not a model, database, or automatic accuracy improvement.

Operate Y4 Point to Memory and Projects.

An explicit memory persists in SQLite and can be retrieved in a later conversation. That is separate from the temporary model context and does not retrain its weights. Internal model-selected tools are read-only and bounded. The development setup used an RTX 5050, Ryzen 7 250, and sixteen gigabytes of RAM; that does not establish a universal performance guarantee.

Operate Y5 Point to source identities, then return to Library and hand over.

Document evidence now includes actual filenames, paths and document identifiers. The display groups excerpts from the same file while preserving their original citation numbers. Gayatri will demonstrate the prepared Lighthouse samples. We will compare the response with its source and save an explicit memory.

Gayatri Rajesh Apotikar

Verbatim script | 04:30-06:30

Read the regular text aloud. Italic operation cues are silent.

Operate G1 D1 Open the sample source at Storage decision.

These are two supplied sample files, not personal documents. They were indexed before this timed demo. The Lighthouse document explicitly says to use SQLite for extracted text, conversations, and project memories. We can inspect that source directly.

Operate G2 D2 Ask the storage question once; use the 10 second limit.

I am asking a question whose answer is written in the sample. Once the response is ready, I compare it with its source card. If it disagrees or abstains, the document remains the evidence we should trust.

Operate G3 D3 Save the explicit memory; show its project scope.

Now I explicitly ask the application to save a demo preference. It appears in this project's Memory view. This save is an intentional database operation, not a claim that the language model has permanently learned the note.

Operate G4 D4 Reload Memory once; stop clicking and hand over.

The note is still listed after the page reloads. Persistence across application restarts is also covered by storage tests. Nimita will now explain what our measured results support and where the prototype still needs improvement.

Use the fallback that matches what happened

Model not ready or response exceeds ten seconds

The local model is not ready within our demonstration window. I will show the indexed source and saved memory instead. Generated answers and persistence are separate functions.

Wrong answer or false abstention

This response needs correction. The sample document explicitly says SQLite. We keep the source reader because a generated response can be wrong even when the file contains the answer.

No source card

This response did not return a usable source card. I will check the storage decision directly in Library.

If the app is unavailable, explain prepared artifacts as prepared. If private content appears, close the view or stop sharing before continuing. Never read private content aloud.

Nimita Jestin

Verbatim script | 06:30-10:00

Read the regular text aloud. Italic operation cues are silent.

Operate N1 Point to retrieval coverage in the results on page 12.

Our evaluation used authored fictional documents, not a representative personal archive. Hybrid retrieval included all required reference facts within its first five passages for ninety-six of ninety-six answerable retrieval questions. This is coverage on a small synthetic corpus, not one hundred percent answer accuracy. Lexical retrieval also reached the same coverage at five.

Operate N2 Point to the historical generation baseline of 0/24.

Historically, both generation conditions matched all requested values on zero of twenty-four answerable questions. The context fitter could discard retrieved evidence after serialization. These are historical measurements. The context repair and later safeguards are implemented in the Linux and Windows versions; the old results remain visible for comparison.

Operate N3 Point to the production repair and direct replay.

The production repair uses the actual serialized UTF-eight size to fit evidence. All sixty-four replay requests retained initial references, stayed within budget and completed without errors. Single-pass RAG matched every requested value on seven of twenty-four answerable questions; the agent did so on three. These were reused questions, not a held-out test or general accuracy. No extra model-selected tool calls occurred.

Operate N4 Show historical summaries and the separate later 9B probe.

The earlier long-document experiment retained ten of twenty-four requested values across three final summaries. It was not rerun. Current question answering uses query-centered excerpts, but bounded context can still omit facts. A later small, post-selected nine-billion-parameter development probe passed eight known-fact checks and declined two unknowns. This is separate from the paper benchmark and does not establish general accuracy.

Operate N5 Show the limits. Do not change model settings.

Current file lists and exact filename checks use index metadata. Greetings now bypass retrieval, avoiding invented project context. Uncited substantive document replies get one retry, including requests such as explain the setup. Failed reference checks show exact excerpts. The local suites passed one hundred sixty-three Linux tests and two hundred Windows-copy tests on Linux, with six native checks skipped. Valid citations still do not prove every claim. Scanned PDFs need external OCR, and the database is not encrypted.

Operate N6 Show future work, then return to the title or overview.

Next steps are stronger long-document coverage, broader held-out evaluation and native Windows GPU testing, which remains unmeasured. Local OCR is another possible extension. These improvements remain proposed. LocalMind combines a working local workflow with measurements of its remaining limits. Thank you to Dr. Sonali Kothari and the panel. We welcome your questions.

Viva questions one to five

The named presenter leads the answer; another teammate can add a verified detail.

1 What problem does the project solve

Answerer Trijal Burman

It helps a person find and reuse information from approved local files and saved project context. It combines search, source reading, explicit memory, and local model assistance. It does not automatically know an entire drive or guarantee every answer.

2 What is the difference between RAG and MCP

Answerer Tiya Singh

RAG retrieves evidence and supplies it to a generator. MCP standardizes how a client discovers and calls tools or reads resources. Our RAG workflow accesses local search and memory through MCP. MCP is not a language model, vector database, or ranking algorithm.

3 Explain the architecture

Answerer Tiya Singh

The browser calls a loopback FastAPI service. Approved files become text and chunks in SQLite. Retrieval selects evidence through the local MCP stdio server and Ollama runs the downloaded model. Greetings bypass that retrieval flow. File inventories use actual index metadata, and source cards group excerpts while keeping citation numbers.

4 Why a custom MCP server instead of only Python function calls

Answerer Tiya Singh

It provides a defined and reusable interface for knowledge and memory functions, including use by compatible local MCP clients. Our implementation uses a real protocol handshake. It adds overhead, so we do not claim MCP automatically improves retrieval accuracy.

5 What persists, and is that model memory

Answerer Gayatri Rajesh Apotikar

SQLite stores extracted text, chunks, conversations, and explicit memories. Plans are stored locally as Markdown. Later requests can retrieve that information. The model context is temporary, and saving records does not alter trained model weights or include every record in every prompt.

Viva questions six to ten

The named presenter leads the answer; another teammate can add a verified detail.

6 Does it have unrestricted access to all local files

Answerer Gayatri Rajesh Apotikar

No. The user approves folders. Unsupported formats, hidden items, known credential filenames, dependencies, links, and the app state folder are excluded. File-size and scan limits also apply. These checks reduce exposure but are not complete secret detection.

7 Did you train or fine-tune the language model

Answerer Nimita Jestin

No. We use existing downloaded Qwen and EmbeddingGemma models. The project concerns the application, retrieval, persistence, tool interface and evaluation. Supplying retrieved context at request time does not train or fine-tune model weights.

8 Which models and hardware did you use

Answerer Tiya Singh

Source defaults are qwen3:4b-instruct-2507-q4_K_M, EmbeddingGemma and 8192 context tokens. This Linux laptop also used the optional qwen3.5:9b-q4_K_M / 12288 profile, with partial CPU offload. Hardware: RTX 5050 with 8 GB VRAM, Ryzen 7 250 and 16 GB RAM. The small development probe is not a universal model ranking; native Windows GPU inference remains unmeasured.

9 Is it completely offline and completely free

Answerer Trijal Burman

Normal use can run offline after the dependencies, runtime, and models are downloaded. There is no required paid hosted API. Hardware, power, storage, and initial downloads still cost resources. A cloud-connected external MCP client could also forward returned contents, so the client must be local to keep the full workflow local.

10 Does 96 out of 96 retrieval coverage mean 100% accuracy

Answerer Nimita Jestin

No. The frozen retrieval experiment put required evidence within five passages on 96 authored questions. Historical generation was 0/24 per method because the old fitter dropped evidence. The repaired production replay reached 7/24 single-pass RAG and 3/24 agent all-value matches on reused questions. These are distinct measurements, not universal answer accuracy.

Viva questions eleven to fifteen

The named presenter leads the answer; another teammate can add a verified detail.

11 What was the context bug, and is the fix deployed

Answerer Nimita Jestin

The old estimate missed JSON escaping and UTF-8 cost, which could discard evidence. The repaired fitter retained references in 64/64 historical replay requests within budget. Current records also preserve exact filenames, paths and document identifiers; oversized metadata does not starve every later fitting record. Greetings bypass retrieval, and substantive document answers pass citation checks. These safeguards do not establish general factual accuracy.

12 Do citations prove factual correctness

Answerer Gayatri Rajesh Apotikar

No. A real citation can still fail to support a claim. Uncited substantive document replies, including imperative requests, can get one tool-free retry. Invalid references fail directly, and failed retries retain safe original excerpts. Grouped cards keep original citation numbers. We still inspect the source passage.

13 Can it summarize every part of a long document

Answerer Nimita Jestin

There is no completeness guarantee. MCP pagination fetches supported extracted text and long summaries process sections, but reduction can lose facts. The historical experiment retained 10/24 requested values across three summaries and was not rerun. Current question answering uses query-centered excerpts instead of fixed prefixes; bounded context can still omit facts.

14 What happens with scanned PDFs or without a model

Answerer Gayatri Rajesh Apotikar

This version does not OCR image-only PDF pages. Text-based PDFs, DOCX, text, Markdown, and supported code can be extracted. Without a chat model, indexing, full-text reading, keyword search, saved memory, and projects still work. Chat labels matching excerpts as extractive mode rather than pretending to generate a summary.

15 What are the main security, Windows, and future-work limits

Answerer Trijal Burman

The database is not encrypted and local access checks do not defeat every threat. Latest suites: 163 Linux tests; 200 Windows-copy tests on Linux with six native checks skipped. The earlier native CI result of 178 tests per interpreter checked a previous commit. Native Windows GPU inference remains unmeasured. Future work includes long-document coverage, held-out evaluation and local OCR.

Results boundaries and final rehearsal

Keep evidence coverage, transport and generated answers separate. Current suites passed 163 Linux tests and 200 Windows-copy tests on Linux, with six native checks skipped. The table separates historical experiments from the later small development probe.

Measurement	Observed result	What to say
Historical retrieval	96/96 at top five	Lexical and hybrid evidence coverage on the authored corpus.
Historical generation	0/24 per method	Original fitter dropped the reference block.
Historical repair replay	RAG 7/24; MCP 3/24	64/64 requests retained evidence. Reused questions; not held-out validation.
Small later 9B probe	8/8 known; 2/2 unknown	Post-selected development checks; not general accuracy. Extra short-summary check failed.
Historical summaries	10/24 fact values	Three documents; not rerun after the repair.

Boundaries to state accurately

Latest metadata checks returned 7 indexed files, 1 exact api.md match and 0 matches for an absent test filename. One hello check took 0.0931 seconds with no model or retrieval call; this is not model-answer latency. No OCR is included, the database is unencrypted, and native Windows GPU inference is unmeasured.

Only if the panel asks for an extension

Only with extra time: open a sample source and choose Summarize in chat, or request a project plan with two milestones and labeled assumptions. Both are reviewable generated drafts. They do not prove full summary coverage or create a finished software project.

Final rehearsal

- Check all four names and PRNs on page 1, then rehearse the handovers in order.
- Keep the total at ten minutes, including two minutes for Gayatri. Leave pauses around the measured results.
- Explain the greeting, metadata and citation repairs without claiming perfect answers or universal model superiority. The earlier Windows CI run checked a previous commit.
- Keep historical research, the small 9B probe and current software checks separate. Next work includes long-document coverage, held-out evaluation, local OCR and Windows GPU testing.